Masaki Allwardt

925-872-1439 | masakiallwardt@berkeley.edu | Pleasanton, CA

Education

UC Berkeley — Berkeley, CA
B.A. Computer Science — *GPA: 4.0.* *Grad. May 2028*

Experience

Capital One — Mclean, VA
Software Engineer - Intern — *Jun 2026 – Aug 2026*

- Reduced manual migration effort by 55% across 22,000+ datasets and increased migration speed 11x by building an AI-assisted data migration platform with a CLI tool and MCP enabled automation, adopted across 3 organizations.
- Saved ∼460 engineering hours by developing an automated remediation tool that detected and corrected mismatched metadata across 6,000+ datasets.
- Reduced build error triage time by ∼40% by building an autonomous agent that monitors Jenkins builds, notifies engineers upon completion, and searches Slack conversations and build history to surface fixes for recurring failures.

Apple — Remote
Machine Learning Engineer - Contract — *Jan 2026 – May 2026*

- Reduced computer vision validation cycle time by 67% across 6 pipeline stages by architecting an agentic validation platform that automated manual QA for large-scale vision training pipelines.
- Reduced end-to-end validation latency by 83% and enabled efficiently processing 10,000+ images/day by designing an LLM-driven orchestration engine with dependency-aware asynchronous execution.
- Enabled rapid deployment of new validation workflows by designing a modular "toolbox" that supports a growing stock of validation filters ranging from image-quality checks to semantic vision classifiers.

Projects

Diplomacy Overseer (RLVR Training Environment) | *PyTorch, OpenEnv, HuggingFace TRL, Docker*

- Awarded $10,000 cash prize in Meta's OpenEnv Hackathon (Multi-Agent Interactions + Scalable Oversight track).
- Designed a scalable oversight environment that trains a 0.5B-parameter model to infer hidden strategies of 7 simultaneous LLM agents from public signals alone: board state, order history, and communication metadata.
- Fine-tuned Qwen2.5-0.5B-Instruct via GRPO over 16,835 training steps across 185 self-play games and ∼17,000 prediction episodes; improved strategy-inference F1 from 0.17 → 0.87 and judge-verified accuracy from 0% → 52.1% using a Claude-powered binary reward judge.

Collab Reference Frames (Wearable-Grounded Correction for Robots) | *SmolVLA, DAgger, LeRobot, Isaac Sim*

- Placed 2nd at the NVIDIA and Nebius Physical AI Sprint Hackathon, shipping a working system within 5 hours.
- Built a correction interface to allow a human working with a SO-101 arm to critique its behavior in natural speech; a VLM grounded the critique in human-egocentric footage from a wearable and rewrote the policy's instruction.
- Trained SmolVLA in Isaac Sim, then closed the sim-to-real gap with 20 rounds of DAgger on the physical arm.

Distributed TRPO Training Engine | *Python, PyTorch, Ray, Gymnasium, CUDA*

- Built a distributed RL training pipeline with 8 parallel rollout workers and a GPU-accelerated PyTorch learner, achieving 2.7× higher environment-step throughput than a single-worker baseline.
- Implemented TRPO from scratch with generalized advantage estimation (GAE), conjugate-gradient optimization using Fisher-vector products, and KL-constrained backtracking line search.
- Benchmarked 1/2/4/8-worker configurations across continuous-control environments and 5 random seeds, reducing wall-clock time to a predefined reward threshold by 26.2%; implemented checkpoint-based recovery from rollout-worker failures.

Technical Skills

Languages: Python, Java, Typescript/JavaScript (Next.js, React), SQL, Dart (Flutter), C, C++, C#
ML & AI: PyTorch, NumPy, Pandas, HuggingFace, TRL, LLMs/VLMs/VLAs, LangGraph, Langchain, vLLM
Backend & Infra: Node.js, FastAPI, PostgreSQL, MongoDB, Docker, AWS, Firebase, Git
DevOps: Docker, Kubernetes, Jenkins, LocalStack, GitHub Enterprise Cloud